

Assignment 7 Report

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1. Examination Registration Form

Code:

```
import tkinter as tk
from tkinter import messagebox

def register():
    if not name_entry.get() or not roll_entry.get() or not course_entry.get()
or not subject_entry.get() or not semester_entry.get() or not
email_entry.get():
        messagebox.showerror("Error", "All fields are required!")
    else:
        messagebox.showinfo("Success", "Registration Successful!")

def clear():
    name_entry.delete(0, tk.END)
    roll_entry.delete(0, tk.END)
    course_entry.delete(0, tk.END)
    subject_entry.delete(0, tk.END)
    semester_entry.delete(0, tk.END)
    email_entry.delete(0, tk.END)

root = tk.Tk()
root.title("Examination Registration Form")

tk.Label(root, text="Full Name:").grid(row=0, column=0, padx=10, pady=5)
name_entry = tk.Entry(root)
name_entry.grid(row=0, column=1, padx=10, pady=5)

tk.Label(root, text="Roll Number:").grid(row=1, column=0, padx=10, pady=5)
roll_entry = tk.Entry(root)
roll_entry.grid(row=1, column=1, padx=10, pady=5)

tk.Label(root, text="Course Name:").grid(row=2, column=0, padx=10, pady=5)
course_entry = tk.Entry(root)
course_entry.grid(row=2, column=1, padx=10, pady=5)

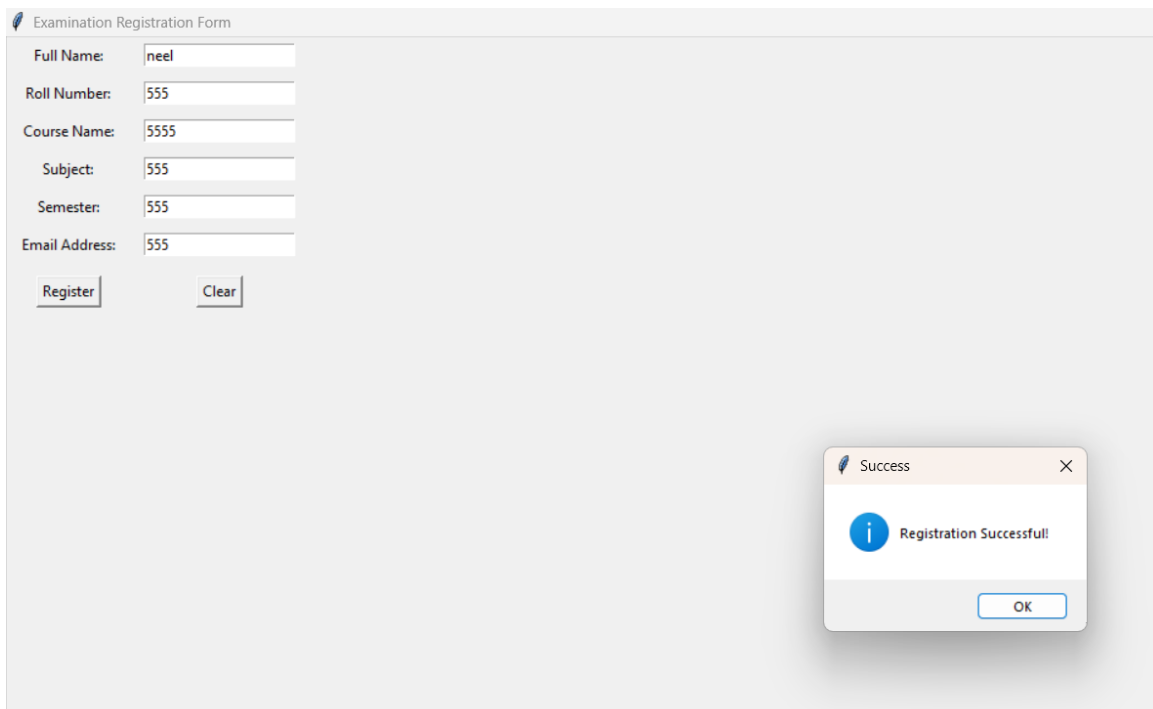
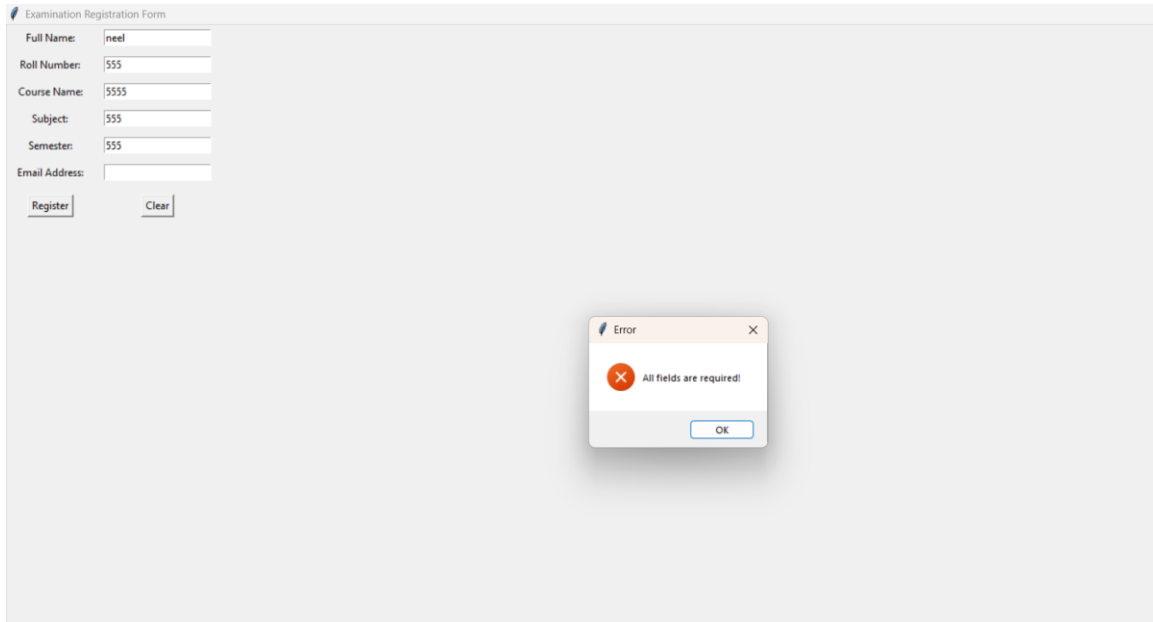
tk.Label(root, text="Subject:").grid(row=3, column=0, padx=10, pady=5)
subject_entry = tk.Entry(root)
subject_entry.grid(row=3, column=1, padx=10, pady=5)

tk.Label(root, text="Semester:").grid(row=4, column=0, padx=10, pady=5)
semester_entry = tk.Entry(root)
semester_entry.grid(row=4, column=1, padx=10, pady=5)
```

```

tk.Label(root, text="Email Address:").grid(row=5, column=0, padx=10, pady=5)
email_entry = tk.Entry(root)
email_entry.grid(row=5, column=1, padx=10, pady=5)
tk.Button(root, text="Register", command=register).grid(row=6, column=0,
pady=10)
tk.Button(root, text="Clear", command=clear).grid(row=6, column=1, pady=10)
root.mainloop()

```



2. Student Record Management System

Code:

```
import tkinter as tk
from tkinter import messagebox

class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def display_result(self):
        if self.marks > 40:
            return "Pass"
        else:
            return "Fail"

def save_record():
    name = name_entry.get()
    marks_str = marks_entry.get()

    try:
        marks = float(marks_str)
        student = Student(name, marks)
        with open("students.txt", "a") as f:
            f.write(f"Name: {student.name}, Marks: {student.marks}, Result: {student.display_result()}\n")
        messagebox.showinfo("Success", "Record saved successfully!")
    except ValueError:
        messagebox.showerror("Error", "Invalid marks input (must be numeric)")
    except Exception as e:
        messagebox.showerror("Error", f"File error: {e}")

def show_result():
    name = name_entry.get()
    marks_str = marks_entry.get()

    try:
        marks = float(marks_str)
        student = Student(name, marks)
        result = student.display_result()
        messagebox.showinfo("Result", f"{name} is {result}")
    except ValueError:
        messagebox.showerror("Error", "Invalid marks input (must be numeric)")

root = tk.Tk()
root.title("Student Record Management System")

tk.Label(root, text="Name").grid(row=0, column=0, padx=10, pady=5)
name_entry = tk.Entry(root)
name_entry.grid(row=0, column=1, padx=10, pady=5)

tk.Label(root, text="Marks").grid(row=1, column=0, padx=10, pady=5)
marks_entry = tk.Entry(root)
marks_entry.grid(row=1, column=1, padx=10, pady=5)

tk.Button(root, text="Save Record", command=save_record).grid(row=2, column=0,
pady=10)
tk.Button(root, text="Show Result", command=show_result).grid(row=2, column=1,
```

```
pady=10)
root.mainloop()
```

The image displays two sequential screenshots of a web application titled "Student Record Management System".

In the top screenshot, the user has entered "neel" for Name and "20" for Marks. The "Show Result" button is highlighted. A modal dialog box titled "Result" is open, displaying a blue information icon and the message "neel is Fail". An "OK" button is at the bottom of the dialog.

In the bottom screenshot, the user has entered "neel" for Name and "56" for Marks. The "Show Result" button is highlighted. A modal dialog box titled "Result" is open, displaying a blue information icon and the message "neel is Pass". An "OK" button is at the bottom of the dialog.

3. Student Information Form

Code:

```
import tkinter as tk
from tkinter import ttk, messagebox
import csv

def submit():
    if not agree_var.get():
        messagebox.showerror("Error", "You must agree to the terms and conditions")
    return

    data = {
        "Name": name_entry.get(),
        "Age": age_scale.get(),
        "Gender": gender_var.get(),
        "Course": course_var.get(),
        "Programming": prog_var.get(),
        "Maths": maths_var.get(),
        "Communication": comm_var.get(),
        "Hobbies": [hobbies_listbox.get(i) for i in hobbies_listbox.curselection()],
        "Comments": comments_text.get("1.0", tk.END).strip()
    }

    with open("student_info.csv", "a", newline='') as f:
        writer = csv.writer(f)
        writer.writerow([data["Name"], data["Age"], data["Gender"],
data["Course"],
                        data["Programming"], data["Maths"],
data["Communication"],
                        ", ".join(data["Hobbies"]), data["Comments"]])
    messagebox.showinfo("Success", "Data saved successfully!")

root = tk.Tk()
root.title("Student Information Form")
root.geometry("400x650")

tk.Label(root, text="Student Information Form", font=("Arial", 14, "bold")).pack(pady=10)

tk.Label(root, text="Name").pack()
name_entry = tk.Entry(root)
name_entry.pack()

tk.Label(root, text="Age").pack()
age_scale = tk.Scale(root, from_=18, to=100, orient=tk.HORIZONTAL)
age_scale.pack()

tk.Label(root, text="Gender").pack()
gender_var = tk.StringVar(value="Male")
tk.Radiobutton(root, text="Male", variable=gender_var, value="Male").pack()
tk.Radiobutton(root, text="Female", variable=gender_var, value="Female").pack()
```

```
tk.Label(root, text="Select Course").pack()
course_var = tk.StringVar(value="Select Course")
course_dropdown = ttk.Combobox(root, textvariable=course_var, values=["BTech",
"MTech", "PhD"])
course_dropdown.pack()

tk.Label(root, text="Select Subjects").pack()
prog_var = tk.BooleanVar()
maths_var = tk.BooleanVar()
comm_var = tk.BooleanVar()
tk.Checkbutton(root, text="Programming", variable=prog_var).pack()
tk.Checkbutton(root, text="Maths", variable=maths_var).pack()
tk.Checkbutton(root, text="Communication", variable=comm_var).pack()

tk.Label(root, text="Select Hobbies").pack()
hobbies_listbox = tk.Listbox(root, selectmode=tk.MULTIPLE, height=4)
for item in ["Reading", "Gaming", "Sports", "Music"]:
    hobbies_listbox.insert(tk.END, item)
hobbies_listbox.pack()

tk.Label(root, text="comments").pack()
comments_text = tk.Text(root, height=4, width=30)
comments_text.pack()

agree_var = tk.BooleanVar()
tk.Checkbutton(root, text="I agree to the term and conditions",
variable=agree_var).pack()

tk.Button(root, text="Submit", command=submit).pack(pady=10)

root.mainloop()
```

Student Information Form

Name
neel

Age
22

Gender
☒ Male
☐ Female

Select Course
MTech

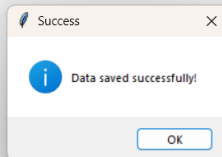
Select Subjects
☐ Programming
☒ Maths
☒ Communication

Select Hobbies
Reading
Gaming
Sports
Music

comments
i am neel prajapati.doing mtec
h at daiict

☒ I agree to the term and conditions

Submit



Student Information Form

Name
neel

Age
22

Gender
☒ Male
☐ Female

Select Course
MTech

Select Subjects
☐ Programming
☐ Maths
☐ Communication

Select Hobbies
Reading
Gaming
Sports
Music

comments
i am neel prajapati.doing mtec
h at daiict

☐ I agree to the term and conditions

Submit



4. Simple Calculator

Code:

```
import tkinter as tk
from tkinter import messagebox

def calculate():
    try:
        num1 = float(entry1.get())
        num2 = float(entry2.get())
        result_label.config(text=f"Result is {num1 + num2}")
    except ValueError:
        messagebox.showerror("Error", "Please enter valid numbers")

root = tk.Tk()
root.title("Simple Calculator")

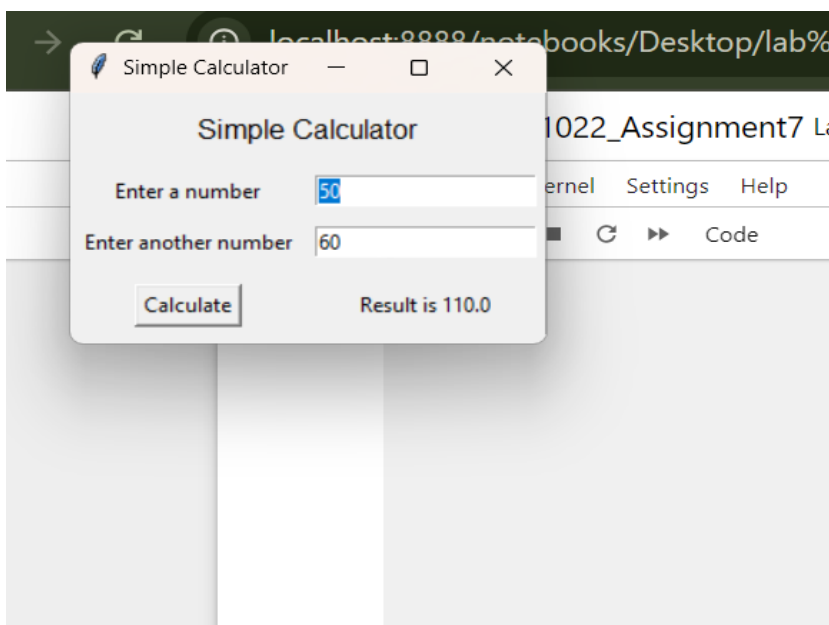
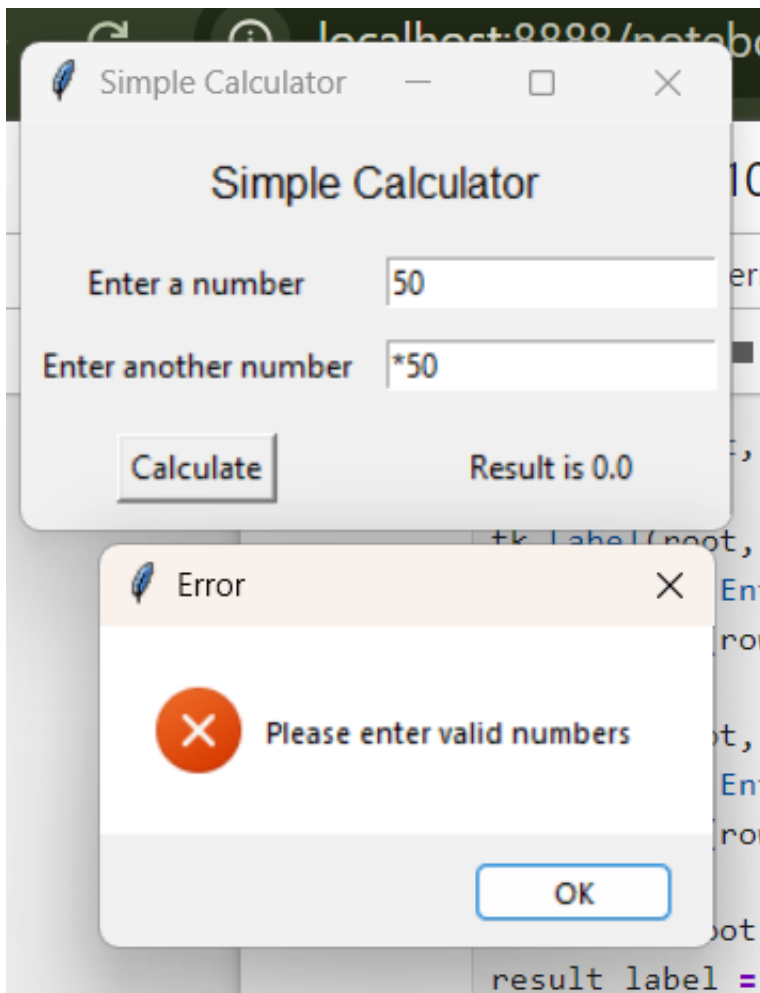
tk.Label(root, text="Simple Calculator", font=("Arial", 12)).grid(row=0,
column=0, columnspan=2, pady=10)

tk.Label(root, text="Enter a number").grid(row=1, column=0, padx=5, pady=5)
entry1 = tk.Entry(root)
entry1.grid(row=1, column=1, padx=5, pady=5)

tk.Label(root, text="Enter another number").grid(row=2, column=0, padx=5,
pady=5)
entry2 = tk.Entry(root)
entry2.grid(row=2, column=1, padx=5, pady=5)

tk.Button(root, text="Calculate", command=calculate).grid(row=3, column=0,
pady=10)
result_label = tk.Label(root, text="")
result_label.grid(row=3, column=1, pady=10)

root.mainloop()
```



5. m to Km converter

Code:

```
import tkinter as tk
from tkinter import messagebox

def convert():
    try:
        meters = float(entry.get())
        km = meters / 1000
        result_label.config(text=f"{km} Km")
    except ValueError:
        messagebox.showerror("Error", "Please enter a valid number")

root = tk.Tk()
root.title("Demo App")
root.geometry("400x200")

tk.Label(root, text="m to Km converter", font=("Arial", 12)).pack(pady=10)

frame = tk.Frame(root)
frame.pack(pady=10)

tk.Label(frame, text="Enter the value in meter").grid(row=0, column=0, padx=5)
entry = tk.Entry(frame)
entry.grid(row=0, column=1, padx=5)
tk.Button(frame, text="Convert", command=convert).grid(row=0, column=2, padx=5)

result_label = tk.Label(root, text="Km")
result_label.pack(pady=10)

root.mainloop()
```

